Psychology 434 Test 1 Study Sheet

Weeks 1–6

1. Start with the causal question

For any causal question, specify the population, intervention, comparison condition, outcome, and timing. If these are vague, the rest of the analysis is unstable.

The average treatment effect is

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)].$$

The conditional average treatment effect is

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

2. Five elementary graph structures

Causality absent:

$$A \coprod B$$

Direct causation:

$$A \to B$$

Fork:

$$A \leftarrow C \to B$$

Chain:

$$A \to C \to B$$

Collider:

$$A \to C \leftarrow B$$

3. Four adjustment rules

Condition on common causes, or sometimes on defensible proxies for them.

Do not condition on mediators if you want the total effect.

Do not condition on colliders.

Treat descendants carefully, because conditioning on them can partly behave like conditioning on their parents.

4. Why model fit is not enough

High R^2, low AIC, or a significant coefficient do not guarantee a good causal estimate.

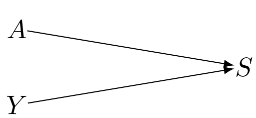
Fit is about prediction of observed data.

Causal workflow is about asking and answering a clearly defined question, stating the causal estimand, making the identification assumptions explicit, and then choosing an estimator that matches that question.

5. Selection bias and measurement bias

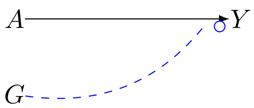
Selection bias can arise through collider conditioning, which mainly threatens internal validity.

Example: suppose treatment A and later distress Y both affect who responds to a follow-up survey S. If we analyse only responders, we condition on a collider.



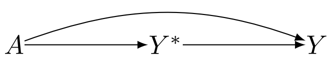
Selection can also threaten transportability if effect modifiers are distributed differently in the analytic sample and the target population.

Example: suppose a learning intervention is randomised, but age group G modifies its effect on self-efficacy Y. If the analytic sample contains mostly younger students while the target population contains more older students, the sample ATE may not transport. The dashed arrow indicates effect modification, not a standard causal path.



Measurement bias can be independent or dependent, and correlated or uncorrelated with the true value. Directed measurement error is especially dangerous because it can open non-causal paths.

Example: suppose treatment A affects true distress Y^*, but it also changes how participants report distress on the recorded measure Y. Then treatment affects measurement of the outcome as well as the outcome itself.



When studying Week 4, do not stop at labels. For each type of bias, practise by inventing at least one example, drawing a DAG, and stating what path is open or blocked.

6. Identification assumptions

Consistency means the intervention is well-defined and links observed outcomes to potential outcomes. If one treatment label hides many substantively different versions, then $Y(a)$ is not clear.

Exchangeability means treated and untreated units are balanced, often after conditioning on measured pre-treatment covariates.

Positivity means both treatment levels occur in every relevant covariate stratum. If one subgroup has only treated or only untreated cases, the effect for that subgroup is not learnable from the data alone.

Without all three, the causal contrast is not identified.

7. Effect modification and interaction

Interaction concerns the joint effect of two interventions.

Effect modification concerns how one intervention effect varies across subgroups.

If you cannot write the target as a potential-outcomes contrast, you have not yet stated the causal estimand clearly.

8. The causal workflow

First define the intervention and the comparison condition.

Then set time zero and define the outcome.

Then draw the DAG and identify the covariates needed for exchangeability.

Then assess consistency, exchangeability, and positivity.

Only after that choose an estimator, such as standardisation, weighting, or regression adjustment.

Design comes before estimation.

9. How to study

Practise with new examples rather than only rereading definitions.

For any scenario, ask:

What is the causal question?

What is the estimand?

What are the likely threats to identification?

What should be measured before time zero?

What should not be conditioned on?